# Nawaz Patel

*Mechanical engineering + everything else*

Pune / Pimpri-Chinchwad, Maharashtra | nawazpatel8008@gmail.com | LinkedIn: linkedin.com/in/nawaz-patel-ab094a271 | GitHub: github.com/nawazjava | ORCID 0009-0006-2164-0287

## About

Mechanical engineering student at PVG COET (Savitribai Phule Pune University, 2024 to 2028) working at the intersection of machine learning, sensing and robotics. I build end-to-end systems, from physics-based simulation and reinforcement learning to hardware prototypes, and turn them into research papers, hackathon entries and prototypes. Aiming for an MSc in Robotics (Oct 2028 intake).

## Research & Publications

### AquaPrecision / SoilSense-RL: Physics-Informed RL for Precision Irrigation

*Target: IEEE Sensors Journal | With R. Naik, S. Jagdale; faculty: Prof. P. Jamdade, Prof. Dr. M. Bhoomkar*

Reinforcement-learning irrigation controller with multi-sensor fusion and vision-language moisture estimation, evaluated across six Indian soil types.

- Built an OpenEnv-style Gym environment from scratch with soil-specific retention for six ICAR soil types, evapotranspiration-driven water balance and region-specific reservoir recharge.
- Trained PPO agents for 400k timesteps per region (100-episode evaluation): survival ranged from 73% (Thar) to 100% (Deccan, Himalayan foothills); ablations showed soil physics and the safety layer to be the most critical components.
- Developing a PINN + MobileNetV2 soil-photo model and an ESP32-based device (moisture, pH, EC, temperature).

### Ultrasonic Radar System using Arduino Uno

*ICAETM 2026, PVG COET (28 to 30 Oct 2026) | Presenting author*

Accepted conference paper on a low-cost ultrasonic radar built on Arduino Uno, co-authored with Prof. Jamdade and Prof. Bhoomkar.

## Engineering & Hackathon Projects

### HAL-IIT Indore AEROTHON 2026: Team Icarus

*With U. Pawar, A. Deshpande, V. Memane*

Lifecycle-robust hybrid-electric propulsion for a fixed-wing UAV (60 kW turboshaft + electric) coupled with a PINN-based digital twin for turbojet health monitoring.

- Full MATLAB simulation chain: generator, rectifier, DC bus, inverter, propeller, energy-management controller, PSO optimisation.
- Baseline design verified feasible with a 26 kg fuel reserve and zero power deficit; two real bugs caught by independent Python cross-validation.
- Research paper drafted targeting IEEE TAES.
- Open-source: MATLAB propulsion-chain code and an offline interactive demo (HTML/SVG/JS) showing endurance and engine/battery power share as the engine ages (illustrative values): github.com/nawazjava/team-icarus-aerothon; live demo.

### Drought-Resilient Crop Watering: OpenEnv Environment

*Scaler x Meta Hugging Face Hackathon | TypeScript, Node, Docker | github.com/nawazjava/hackathons-meta*

OpenEnv-compliant simulated irrigation environment where AI agents manage soil moisture across a crop grid under a limited water reservoir and 24-hour weather forecasts.

- Agent actions WATER(crop, amount) and WAIT(hours); standard step / reset / state interface with typed models and programmatic graders.
- Three graded tasks: small garden, commercial field with variable weather, and a drought crisis with frequent heatwaves and scarce water.
- Includes a baseline inference script, Dockerfile and ESP32 firmware folder.

### Pathology: Gemini-Powered Web App (prototype)

*TypeScript, Vite, Google AI Studio / Gemini API* | *github.com/nawazjava/pathology-*

Early-stage web app prototype for pathology-related work, built on the Google AI Studio template with the Gemini API.

### Smart India Hackathon 2026: Team Yantras (Team Leader)

*Nominated by PVG COET* | *Manuscript prepared for IJAEOG (corresponding author)*

Led a five-member mechanical + computer science team on Bhu Kavach, a real-time mine-subsidence early-warning system (SIH 26025). github.com/nawazjava/sih-subsidence

- End-to-end stack: ESP32 + LoRa sensor nodes (tilt, strain, vibration) and simulator over MQTT, FastAPI backend with WebSocket live feed, Random Forest risk scoring (Safe / Warning / Critical), React dashboard, Telegram alerts; deployed on Render and Vercel.
- On simulated runs the Random Forest reached macro-F1 0.95 vs 0.88 for fixed-threshold rules; trained and tested on simulated data only, field retraining pending.

### Formula Bharat EV Accumulator Thermal Simulation

*MATLAB / Simulink*

Thermal model of a 22s10p accumulator (four modules, Molicel P50B cells) with forced-air cooling.

### Pune Metro Planner: Interactive Network Designer

*JavaScript web app* | *GitHub* | *live demo*

Browser tool for designing a transit network on a live map: place stations, draw routes and watch the shortest path recompute as you edit.

- Dijkstra and A* pathfinding, animated trains on the Purple, Aqua and Orange lines, and a lawnmower area-coverage sweep generator (survey-drone / robot-mower style paths).
- Save/load networks, share links and JSON export.

### Paper Reproduction Toolkit

*Python, scikit-learn*

Reusable workflow and test suite (sanity, unit, integration, comparison to scikit-learn) for reproducing ML papers, with an AlexNet reproduction plan.

## Leadership & Experience

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| **AIESEC Pune** | Organising Committee Vice President (Dec 2025 to present); OC Member, OLC 2025; Junior Manager, Corporate Sector |
| **Robotics Club, PVG COET** | Co-organiser, ROBOVERSE 2026 and PBL Quest 2026; built a 177-company tiered sponsor database and outreach plan |
| **EDC PVG** | Finance and Sponsorship team member (Sep 2025 to present) |
| **Mood Indigo, IIT Bombay** | Indigo Squad member (Dec 2024) |
| **Freelance** | Copywriter (Jul 2024) |

## Education

**B.E. Mechanical Engineering**, PVG College of Engineering & Technology, SPPU, Sep 2024 to 2028
HSC (Science), Balbhim Arts Science Commerce College, Beed, 2022 to 2024

## Skills

**ML & Data:** Machine learning, Random Forest, reinforcement learning (PPO), PINNs, scikit-learn, PyTorch-style CNNs; certified in Intro to Machine Learning
**Engineering:** MATLAB / Simulink, Python, Arduino and ESP32 prototyping, thermal and propulsion modelling
**Creative:** Music production, video editing, cinematography, photography, graphic design